Instructions (guidelines and rules) for the elaboration of the MOS Project
(P3 of 2025/2026: course duration 7 weeks)

Instructors:

- José Rui Figueira
- Sérgio Batista

A. Student group

A group is composed of three to four (preferably) students. Please form your group during Week 1.

B. Deadlines

a. Project guidelines and rules: February 18, 2026.
b. Group registration on Fénix: March 04, 2026 (after this deadline no further group changes are allowed)
c. Communication (by e-mail) of the group Project topic with a brief description to Prof. Sérgio Batista (sergio.batista@tecnico.ulisboa.pt) and Prof. José Rui Figueira (figueira@tecnico.ulisboa.pt): March 04, 2026
d. Project electronic submission on Fénix by April 07, 2026 at 11:59 pm
e. Presentation (Week 8): April 09, 2026 – to be confirmed (the presentation may consist only of questions related with the project).

C. Project upload instructions

1. Electronic submission via Fenix:
   a. Zip file with the name as follows: Group#_2026.zip, where # represents your group number attributed by Fénix.
   b. In the zip file, please add your Project in pdf format (Number of pages: at most 20) with the name following the rules in a), e.g., PDF_FigueiraEtAl_MOS_2025.pdf.
   c. Add your talk in pdf format (Number of slides: at most 20), with the name following the rules in a), e.g., PPT_FigueiraEtAl_MOS_2025.pdf.
   d. Add also the Cplex/Gurobi, Excel, and/or other files (with no specific names) related with the Solver solution and data.
   e. Finally, in the first page of your project and presentation add the students' identifications and contacts.
2. For the submission do the following steps (Enter in CIO webpage):
   a. Select “Avaliações”
   b. Select “Projectos”
   c. Then, you have “Project_MOS_2025”
   d. Attention! Create/check an “individual group” just for your group.

(Additional request: Please do not make multiple submissions. Submit your project only when you are sure it corresponds to the last version).

D. Project contents and structure:

1. Cover with students' identifications and contacts.
2. Statement of contribution and AI usage (see Lecture 1 slides) – does not towards the the final page count.
3. Executive summary (maximum of one page).
4. Structure
    a. Introduction
    b. Decision situation and context characterization
    c. Problem definition
    d. MILP model specifications
    e. Software model
    f. Data (real or constructed simulated data)
    g. Results and robustness/sensitivity/stability analysis
    h. Conclusions and Recommendations
    i. References

    (This structure is not compulsory; you may define a different one.)
5. Number of pages: at most 20.

E. Assessment criteria:

1. Relevance of the chosen topic.
2. Presentation and project structure.
3. Quality of resolution, wording, and clarity of explanations provided.
4. Quality of the solver code and computer solution.
5. Autonomy/independence demonstrated during the elaboration of the project.
6. References used.
7. If necessary, a student may be asked to take an oral exam by videoconference (see quality control).

(Additional observation: The projects will be classified in different groups: excellent, very good, good, average, below average. All the projects are graded with respect to the best ones.)

F. Notes:

1. Students with similar Projects will be penalized.
2. Projects inspired on examples from previous years' projects/exams, from the "MOS Lecture Notes and Slides", or from classes notes cannot be used.
3. If the topic of the project is taken from a book, internet, etc., the sources must be mentioned (otherwise, you will have a penalty).
4. Topics or innovative or more complex problems will be rewarded.
5. During the project period, specific questions about the problems selected by the students will not be answered. However, questions about the subjects taught in the classes can be answered.

(Additional note: Screening software for detecting overlapping with sources will be used.)

G. Quality control:

1. After the publication of grades, students can be selected to be submitted to an oral exam (Attention: By default, all the students are selected, and they can be submitted to an oral exam immediately after the publication of the project grades).
2. For those who will consult/check their project, they can also be submitted to an oral exam.

H. Suggested topics

1. Epidemic logistics
2. Drones trajectory planning
3. Mining exploration
4. Political districting
5. Social networks
6. Smart cities
7. Space debris removal
8. Berth allocation problem
9. Family TSP (Scattered storage in warehouses)
10. Just-in-time scheduling/routing
11. Telecommunication networks design
12. Hub-location/p-median/Hamiltonian p-median
13. Traveling purchaser problem/Traveling thief problem
14. Thesis/course scheduling
15. Exam timetabling
16. Cutting and/or packing problems
17. Routing/scheduling problems with time dependent features (e.g., service time, traveling time)
18. The RNA-Folding problem
19. TSP and genomics
20. Molecular sequence analysis
21. Tanglegrams and coevolution
22. Infeasible linear systems and multicriteria analysis
23. Location of primary schools
24. Location of healthcare facilities
25. Electric Vehicle Routing Problem (EVRP)
26. Disaster Relief Logistics
27. Bike-Sharing Rebalancing Problem
28. Operating Room Scheduling
29. Crew Scheduling (Airline or Railway)

I. Sources (examples): institutes, national OR societies, OR groups

1. INFORMS: https://www.informs.org/
2. IFORS: https://www.ifors.org/
3. EURO: https://www.euro-online.org

4. APDIO: http://apdio.pt/home
5. ROADEF: https://www.roadef.org
6. OR Society: https://www.theorsociety.com/
7. SEIO: http://www.seio.es/
8. AIRO: https://www.airo.org
9. GOR: https://www.gor-ev.de/

J. Sources (examples) : Newsletters

1. https://pubsonline.informs.org/magazine/orms-today
2. https://www.optima-bot.org/index.php/en/2-non-categorizzato/36-optima-newsletter-en-en

K. Sources (examples): books containing applications

1. Kulkarni, A.J. and Satapathy, S.C. (Editors) (2020). *Optimization in Machine Learning and Applications*. Springer Nature, Singapore.
2. Gusfield, G. (2019). *Integer Linear Programming in Computational and Systems Biology*. Cambridge University Press, Cambridge, UK.
3. Mansini, R, Ogryczak, W., and Speranza, M.G. (2015). *Linear and Mixed Integer Programming for Portfolio Optimization*. Springer International Publishing Heidelberg Germany.
4. Murthy, G. (Editor) (2015). *Applications of Operations Research and Management Science: Case Studies*. Springer Science+Business Media, New York, NY, USA.
5. Murthy, K. (Editor) (2015). *Case Studies in Operations Research Applications of Optimal Decision Making*. Springer Science+Business Media, New York, NY, USA.
6. Panik, M.J. (2019). *Linear Programming and Resource Allocation Modeling*. John Wiley & Sons, Inc., Hoboken, NJ, USA.
7. Plà-Aragone, L.M. (Editor) (2015). *Handbook of Operations Research in Agriculture and the Agri-Food Industry*. Springer Science+Business Media, New York, NY, USA.
8. Poler, R. & Mula, J. & Díaz-Madroñero, M. (Editors) (2014). *Operations Research Problems: Statements and Solutions*. Springer Science+Business Media, New York, NY, USA.
9. Ravindran, A. R. (Editor) (2009). *Operations Research Applications*. CRC Press, New York, NY, USA.
10. Stadler, W. (Editor). (1998). *Multicriteria Optimization in Engineering and in the Sciences*. Springer Science + Media LLC, New York, NY, USA.

L. Sources (examples): articles (some are probably not accessible at IST).

1. ScienceDirect: https://www.sciencedirect.com/
2. JSTOR: https://www.jstor.org/
3. INFORMS pubs: https://pubsonline.informs.org/